

SHIVARAMAN T

Salem, Tamil Nadu, India

intelshivaram005@gmail.com | +91-90426-56366 | linkedin.com/in/shivaramant | github.com/SHIVARAMAN15

Profile Summary

B.Tech ECE (AI & Cybernetics) student with a strong interest in the **Hardware and VLSI domain**. Strong foundation in **digital electronics, CMOS basics, embedded platforms, microcontrollers, SoC fundamentals, and RTL concepts**, along with knowledge of **IoT systems, embedded security, cybersecurity fundamentals, and AI-driven embedded solutions**. Experienced in **hardware integration, firmware development, secure data handling, and system-level debugging**.

Education

Vellore Institute of Technology

Sept 2023 – May 2027

B.Tech in Electronics and Communication Engineering (AI & Cybernetics)

CGPA: 8.66 / 10

Technical Skills

Programming: C++, Embedded C, Python

Hardware & Embedded: ESP32, Arduino, Sensors, Relays, Firmware Development, Hardware Debugging

Core Domains: IoT Systems, Embedded Security, Cybersecurity Fundamentals, AI/ML Fundamentals, Edge Computing

Cybersecurity: Secure Communication, Basic Encryption, Data Protection

VLSI & Core Electronics: Digital Electronics, CMOS Basics, SoC Fundamentals, RTL Design Concepts

Signal Processing: Signals and Systems, Filtering, ADC Fundamentals

Tools: MATLAB, Simulink, Proteus, LTspice, Git

Experience

Aqualis Inspection Services, Dubai — Engineering Intern

May 2025 – Jun 2025

- Contributed to inspection and validation of **5+ embedded monitoring systems** involving real-time sensor data.
- Performed testing, documentation, and **fault analysis** to improve reliability and safety compliance.
- Supported **system debugging and secure data handling** in industrial environments.

Projects

Smart Aeroponics System — Embedded Hardware Project

GitHub

- Developed an ESP32-based platform for **real-time IoT monitoring** of environmental parameters.
- Integrated pH, TDS, temperature, and water-level sensors with **reliable data acquisition**.
- Implemented a **closed-loop control system** for automated nutrient dosing and misting cycles.
- Improved system efficiency through **optimized embedded control logic**.

Automatic Aerator System (AquaDox)

GitHub

- Designed an **IoT-based monitoring system** for dissolved oxygen optimization.
- Applied **threshold-based automation** for intelligent aeration control.
- Enhanced firmware to improve **energy efficiency and system performance**.
- Performed **sensor calibration and signal conditioning** for stable operation.

Smart Patrol Bot — Autonomous Surveillance Robot

GitHub

- Built an autonomous robot with **AI-based surveillance and motion detection**.
- Enabled **real-time video streaming and alert-based communication**.
- Implemented motor control and **reliable embedded communication systems**.
- Utilized **edge-based processing** for efficient response.

Publications

Smart Helmet Systems: IoT Enhanced Safety and Integrated Display for Two Wheelers

ICRAME 2025, NIT Silchar

Achievements

- Global Rank 5135 — TCS CodeVita Season 13
- Top 100 — AWS Machine Learning Challenge
- Selected Participant — ISRO Bharatiya Antariksh Hackathon 2024

Certifications

Arduino Programming & Embedded Systems • MATLAB (Signal Processing) • Embedded Systems